Project Planning Phase (Logic)

Agile Sprint Planning and Velocity Estimation

Agile Framework Overview

The development of the Automated Network Request Management System follows the Agile Software Development methodology. Instead of completing the entire project in a single phase, the work is divided into multiple development iterations called Sprints. Each sprint is designed to deliver a set of functional features that can be implemented, tested, and reviewed independently. This iterative approach improves project flexibility, enables continuous feedback, and ensures the solution evolves based on project requirements.

To organize the development process efficiently, the project is structured using Epics, User Stories, and Story Points, which help estimate development effort and track overall project progress.

Agile Planning Definitions

- **Sprint:** A Sprint represents a fixed development cycle of two weeks, during which a predefined set of project tasks is completed. Each sprint delivers a functional module that contributes to the final application.
- **Epic:** An Epic is a high-level functional module that represents a major feature of the project. Since an Epic is usually too large to complete within a single sprint, it is divided into smaller development activities called User Stories.
- **User Story (US):** A User Story represents an individual functional requirement that delivers a specific feature to the end user. Each user story focuses on implementing one measurable functionality within the application.
- **Story Point:** Story Points are used to estimate the development effort required for each user story. The estimation considers implementation complexity, technical challenges, and development time.

Story Point Estimation Scale

The following scale was used while estimating development effort:

- **1 Story Point** – Very simple configuration tasks such as catalog creation or basic interface customization.
- **2 Story Points** – Standard development activities including form design and simple business logic implementation.
- **3 Story Points** – Moderately complex tasks involving UI Policies, validation rules, and dynamic field behavior.
- **5 Story Points** – Highly complex implementation involving workflow automation, backend integration, reporting dashboards, and database operations.

Sprint 1 – User Interface Development & Request Management

The first sprint concentrates on building the user-facing components of the application. The objective is to provide employees with an intuitive Service Portal interface that simplifies the process of submitting network service requests while ensuring complete and accurate information is collected.

Epic 1 – Network Request Portal (FR-1)

The first Epic focuses on creating the Network Request Catalog within the Service Portal.

US-1 – Employee Profile Retrieval

The application retrieves the logged-in employee's information automatically using ServiceNow session data. This eliminates repetitive manual data entry and improves the accuracy of submitted requests.

Story Points: 2

US-2 – Network Request Catalog Creation

A dedicated Service Catalog Item is developed to allow employees to submit network-related requests through a centralized self-service portal.

Story Points: 1

Epic 2 – Dynamic User Interface (FR-2)

This Epic improves the usability of the request form by displaying only the information required for the selected request type.

US-3 – Dynamic Field Visibility

UI Policies are configured to display relocation-specific fields only when employees choose the **Relocation** request type.

Story Points: 3

US-4 – Catalog Variable Organization

Network request variables are grouped and arranged to improve readability and simplify the request submission process.

Story Points: 3

US-5 – Mandatory Field Validation

Validation rules ensure that users cannot submit incomplete requests by enforcing mandatory field checks.

Story Points: 3

Sprint 1 Summary

The total development effort for Sprint 1 is **12 Story Points**. This sprint establishes the user interface, request submission process, and input validation required for the application.

Sprint 2 – Workflow Automation & Data Management

The second sprint focuses on automating the backend processing of submitted requests. This includes manager approvals, workflow execution, database integration, and reporting.

Epic 3 – Approval Workflow Automation (FR-3)

This module automates the complete approval process using ServiceNow Flow Designer.

US-6 – Manager Approval Workflow

The submitted request is automatically routed to the reporting manager for approval.

Story Points: 2

US-7 – Notification Configuration

Automatic email and portal notifications are generated to keep both employees and managers informed about request status.

Story Points: 2

US-8 – Request Status Management

The application updates the request status dynamically throughout the workflow, displaying values such as **Requested**, **Approved**, and **Rejected**.

Story Points: 2

US-9 – Workflow Validation

Additional workflow conditions are implemented to ensure that only approved requests proceed to the fulfillment stage.

Story Points: 4

Epic 4 – Database Integration (FR-4)

This module focuses on securely storing approved network requests.

US-10 – Database Record Creation

Once manager approval is received, the validated request details are automatically stored in the **u_network_database** table, eliminating manual record maintenance.

Story Points: 5

Epic 5 – Monitoring & Reporting

The final Epic of Sprint 2 focuses on monitoring application performance.

US-11 – Dashboard Development

Interactive dashboards are created to display request statistics, approval summaries, and network provisioning activities for administrators.

Story Points: 5

Sprint 2 Summary

The total estimated effort for Sprint 2 is **20 Story Points**. This sprint completes the automation layer, database integration, and reporting functionalities of the application.

Development Velocity Calculation

The overall development effort is estimated by combining the story points completed in both sprints.

- **Sprint 1 Total:** 12 Story Points
- **Sprint 2 Total:** 20 Story Points

Overall Story Points Completed: 32

Number of Sprints: 2

The development velocity is calculated using the Agile formula:

Velocity = Total Story Points Completed / Number of Sprints

Velocity = 32 / 2 = 16 Story Points per Sprint

Conclusion

Based on the completed sprint planning activities, the estimated development velocity for the **Automated Network Request Management System** is **16 Story Points per Sprint**. This estimation provides a practical measure of the project's development capacity and serves as a baseline for planning future enhancements, testing activities, deployment tasks, and additional ServiceNow features. It also helps ensure that project deliverables are completed within the planned schedule while maintaining quality and consistency.